Mohamed Makhlouf

+966 533741162 | muhamedomar.g@gmail.com | linkedin.com/in/mhd3omar | github.com/Mhdomer | Blog

Professional Summary

Computer Science student (Networking & Security) specializing in Cloud Security, DevSecOps, and Automated Compliance. **3rd place, ASEAN Cyber Race 2025 CTF finals** (15 teams, 8 countries). Hands-on architecting production AWS environments with Terraform, engineering event-driven remediation, and enforcing CI/CD security gating with Trivy, SonarQube, and Checkov. Dean's List (3.43 GPA), active Saudi Iqama, seeking a Cloud / Security internship: **Sept 2026 - Feb 2027**.

Education

Universiti Teknologi Malaysia — Johor, Malaysia

B.Sc. in Computer Science (Networking and Security) — *Oct 2023 - Oct 2027*

- **Academic Standing:** CGPA 3.43/4.00 (Dean's List)
- **Honors:** 3rd Place, ASEAN Cyber Race 2025 CTF Finals (15 teams from 8 countries), Cisco Networking Academy (Nov 2025); AWS Agentic AI Hackathon, LEAP 2026 Riyadh (Sep 2026)

Technical Skills

Cloud & Infrastructure: AWS (VPC, EC2, S3, IAM, EventBridge, Lambda, NAT Gateway), Terraform (IaC), Serverless, Boto3, CloudWatch, CloudTrail

DevOps & CI/CD: GitHub Actions, Docker, Kubernetes, Ansible, Linux (RHEL/Ubuntu), Bash, Python, Git, YAML

Cloud Security & Compliance: IAM Policy Design, Least Privilege, Trivy (Container Scanning), SonarQube (SAST/SCA), Checkov (IaC Security), CIS Benchmarks

Networking & Systems: TCP/IP, OSPF, VLANs (802.1Q), IPSec VPN, Cisco IOS, Access Control Lists (ACLs), Wireshark, SQL

Certifications

Held: Cisco CCST: Cybersecurity (Jan 2026); Cisco CCNA: ITN, SRWE, ENSA (Mar 2026); Cisco DevNet Associate (Aug 2026); TryHackMe DevSecOps & Security Engineer Paths (Apr 2026)

In progress: AWS Solutions Architect Associate (SAA-C03); Red Hat Certified System Administrator (RHCSA) (both expected Q4 2026)

Projects

Automated Cloud Compliance & Remediation Engine | GitHub | *AWS, Lambda, Python* — Apr 2026 - Sep 2026

- Engineered a serverless remediation engine on EventBridge and Lambda that detects and fixes **public S3 ACLs, weak KMS encryption, unencrypted EBS volumes, and open ports (22/3389)** on the triggering CloudTrail event, not at the next audit cycle.
- Hardened it against real failure modes (botocore adaptive rate-limiting, self-invocation loop prevention, and SQS DLQ routing), backed by **325 Python unit tests** and **25 offline Terraform tests**.
- Built a **read-only MCP server** exposing 6 inspection tools for natural-language posture and log audits, enforcing read-only boundaries via **Python AST analysis** and a least-privilege IAM reader policy.

MindCraft: Three-Tier Web Infrastructure & Gated CI/CD | GitHub | *AWS, Docker, CI/CD* — Oct 2025

- Provisioned a self-managed multi-tier AWS environment via Terraform (**47 resources** across VPC, ALB, EC2, and CloudWatch) with isolated subnets and least-privilege IAM roles.
- Engineered a GitHub Actions CI/CD pipeline enforcing shift-left security: **Trivy image scanning that blocks critical CVEs**, with automated deployment to EC2 via SSM.

Cloud-Native Clinic Management System (PDMS) | GitHub | *AWS, Terraform, IAM* — Apr 2026 - Present

- Architecting a HIPAA-aligned three-tier AWS infrastructure with isolated private DB subnets, NAT Gateways, and strict security groups, validated via **Checkov static IaC security analysis**.
- Integrated CloudTrail audit logging and CloudWatch metric alarms to establish centralized observability, audit readiness, and incident visibility across all tiers.